

FULL-STACK GYM MANAGEMENT SYSTEM

Brief Description

This is a robust, event-driven, full-stack platform designed to streamline and modernize daily fitness center operations. Leveraging an advanced Next.js/React frontend and a highly secure Spring Boot backend, the system offers a user-friendly digital environment for gym administrators, trainers, and members. It automates critical workflows including subscription management, attendance tracking, secure payment processing, and role-based data isolation. By integrating real-time notifications and scalable background processing, the platform ensures high operational efficiency and fosters elevated member engagement.

Principle

- **Event-Driven Architecture:** Utilizes continuous background event processors to trigger real-time notifications, payment reminders, and systemic audit logs asynchronously without blocking primary user workflows.
- **Role-Based Access Control (RBAC):** Adopts enterprise-grade security protocols utilizing JSON Web Tokens (JWT) and 2FA, enforcing strict data boundaries so Super Admins, Trainers, and Members access only authorized information.
- **Stateless Scalability:** Engineered with a decoupled React/Spring Boot microservice-inspired architecture, allowing the API and User Interface to be independently scaled, maintained, and deployed on modern edge networks.
- **Data-Driven Analytics:** Processes raw operational metrics (attendance, revenue, active subscriptions) into comprehensive visual dashboards, empowering decision-makers with actionable insights.

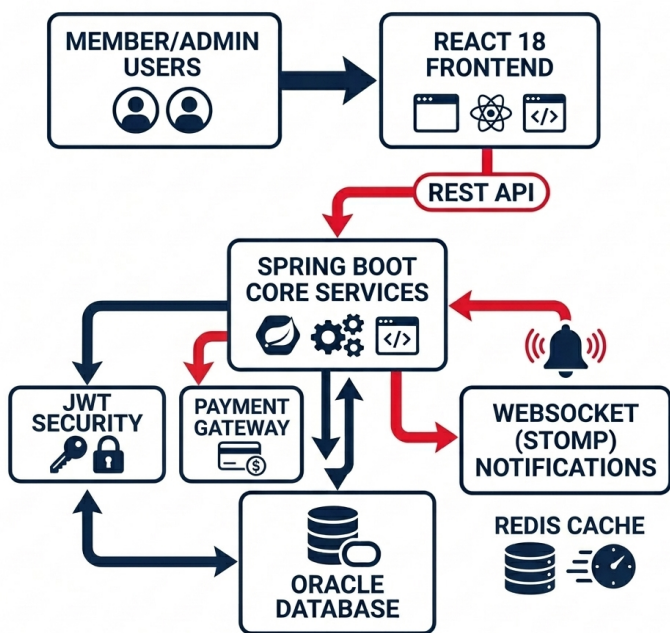
Beneficiaries

- Center Owners & Operators
- Admins and Front-desk Staff
- Personal Trainers & Coaches
- Gym Members & Subscribers
- Financial Auditors & Maintainers

Benefits

- Significantly Reduced Administrative Overhead
- Automated Subscription & Revenue Tracking
- Enhanced Transparency via Tamper-proof Audit Logs
- Improved Stakeholder Communication (Real-time)
- Scalable Architecture for Multi-Branch Expansion

System Architecture



Enterprise Architectural & Security Highlights

High-Performance Implementation

- **Rapid Client UI:** Built with React 18, TypeScript, and functional components ensuring seamless engagement.
- **Backend Engine:** Secured REST API on Java 17 / Spring Boot 3 handling operations with under 200ms latency.
- **Data Integrity:** Optimised, ACID-compliant data persistence via Oracle Database using strictly parameterised SQL.

Security & Real-time Functional Modules

- **Zero-Trust Security:** Full OWASP compliance via BCrypt hashing, JWT Auth, and Email OTP 2-Factor Authentication.
- **Advanced RBAC:** Multi-tier dynamic isolation strictly governing data interfaces for Admins, Owners, Trainers, and Members.
- **Low-Latency Messaging:** Bi-directional WebSockets (STOMP Protocol) facilitating critical alerts and live chat seamlessly.

Figure 1: Full-Stack IT Architecture & Core Services

Functional Modules

- Multi-Role Dashboard System
- Membership Lifecycle Management
- Attendance & Progress Tracking
- Subscription & Payment Workflows
- Trainer-Member Live Chat
- Super Admin Audit Logging

Future Scope

- AI-Driven Workout Plans
- Biometric Login Integration
- Mobile App (React Native)
- Multi-Branch Franchise Mode
- Payment Gateway (Stripe)
- Wearable Device Sync
- Offline Mode & Caching

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